

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.7171]
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C:\Admin>javac server.java

C:\Admin>javac client.java

C:\Admin>java server
Server started...
Client: /127.0.0.1
```

```
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C:\Admin>java client
Enter number (or 'exit'): 5
Server says: 5 is prime.
Enter number (or 'exit'): 4
Server says: 4 is not prime.
Enter number (or 'exit'): 6
Server says: 6 is not prime.
Enter number (or 'exit'): |
```

```
client.java server.java
File Edit View

import java.io.*;
import java.net.*;

public class server {
    public static boolean isPrime(int n) {
        if (n <= 1) return false;
        for (int i = 2; i <= Math.sqrt(n); i++)
            if (n % i == 0) return false;
        return true;
    }
    public static void main(String[] args) {
        try {
            ServerSocket ss = new ServerSocket(12345)
        } {
            System.out.println("Server started...");
            Socket s = ss.accept();
            System.out.println("Client: " + s.getInetAddress());
            BufferedReader in = new BufferedReader(
                new InputStreamReader(s.getInputStream()));
            PrintWriter out = new PrintWriter(
                s.getOutputStream(), true);
            String line;
            while ((line = in.readLine()) != null &&
                !line.equalsIgnoreCase("exit")) {
                int num = Integer.parseInt(line);
                out.println(
                    isPrime(num) ?
                    num + " is prime." :
                    num + " is not prime."
                );
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```
client.java server.java
File Edit View

import java.io.*;
import java.net.*;
import java.util.Scanner;

public class client {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("127.0.0.1", 12345);
            BufferedReader in = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));
            PrintWriter out = new PrintWriter(
                socket.getOutputStream(), true);
            Scanner sc = new Scanner(System.in)
        ) {
            while (true) {
                System.out.print("Enter number (or 'exit'): ");
                String msg = sc.nextLine();
                out.println(msg);
                if (msg.equalsIgnoreCase("exit"))
                    break;
                String response = in.readLine();
                System.out.println("Server says: " + response);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

PRAC : 01

```
C:\Windows\System32\cmd.e  X  +  v

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C:\Admin>javac server1.java

C:\Admin>java server1
Client: hey!
Server: Hello
```

```
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C:\Admin>javac client1.java

C:\Admin>java client1
Client: hey!
Server: Hello
Client: |
```

```
client1.java  X  server1.java

File  Edit  View

import java.io.*;
import java.net.*;
import java.util.*;

public class client1 {
    public static void main(String[] a) throws Exception {
        Socket s = new Socket("localhost", 1234);
        BufferedReader in = new BufferedReader(
            new InputStreamReader(s.getInputStream()));
        PrintWriter out = new PrintWriter(
            s.getOutputStream(), true);
        Scanner sc = new Scanner(System.in);
        String msg;
        while (true) {
            System.out.print("Client: ");
            msg = sc.nextLine();
            out.println(msg);
            if (msg.equals("exit"))
                break;
            System.out.println("Server: " + (msg = in.readLine()));
            if (msg.equals("exit"))
                break;
        }
        s.close();
    }
}
```

```
client1.java  server1.java

File  Edit  View

import java.io.*;
import java.net.*;
import java.util.*;

public class server1 {
    public static void main(String[] a) throws Exception {
        ServerSocket ss = new ServerSocket(1234);
        Socket s = ss.accept();
        BufferedReader in = new BufferedReader(
            new InputStreamReader(s.getInputStream()));
        PrintWriter out = new PrintWriter(
            s.getOutputStream(), true);
        Scanner sc = new Scanner(System.in);
        String msg;
        while (true) {
            System.out.println("Client: " + (msg = in.readLine()));
            if (msg.equals("exit"))
                break;
            System.out.print("Server: ");
            msg = sc.nextLine();
            out.println(msg);
            if (msg.equals("exit"))
                break;
        }
        s.close();
        ss.close();
    }
}
```

PRAC : 02

```
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C:\Admin>javac server2.java

C:\Admin>java server2
Server started...

C:\Admin>
```

```
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C:\Admin>javac client2.java

C:\Admin>java client2
Enter a number: 7
Result from server: ODD

C:\Admin>
```

```
client2.java  X server2.java

File Edit View

import java.net.*;
import java.io.*;
import java.util.Scanner;

public class client2 {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("localhost", 5000);
            BufferedReader in = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));
            PrintWriter out = new PrintWriter(
                socket.getOutputStream(), true);
            Scanner sc = new Scanner(System.in);
            System.out.print("Enter a number: ");
            int num = sc.nextInt();
            out.println(num);
            String result = in.readLine();
            System.out.println("Result from server: " + result);
            socket.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```
client2.java  server2.java

File Edit View

import java.net.*;
import java.io.*;

public class server2 {
    public static void main(String[] args) {
        try {
            ServerSocket server = new ServerSocket(5000);
            System.out.println("Server started...");
            Socket client = server.accept();
            BufferedReader in = new BufferedReader(
                new InputStreamReader(client.getInputStream()));
            PrintWriter out = new PrintWriter(
                client.getOutputStream(), true);
            int number = Integer.parseInt(in.readLine());
            if (number % 2 == 0)
                out.println("EVEN");
            else
                out.println("ODD");
            server.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

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```

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C:\Admin>javac server3.java

C:\Admin>java server3

C:\Admin>

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C:\Admin>javac client3.java

C:\Admin>java client3
Enter first number: 2
Enter second number: 3
Enter operator (+ - * /): -
Answer = -1

C:\Admin>

```

```

server3.java  client3.java  ×  +
File Edit View

import java.io.*;
import java.net.*;
import java.util.*;

class client3 {
public static void main(String[] args) throws Exception {
Socket s = new Socket("localhost", 1234);
Scanner sc = new Scanner(System.in);
DataOutputStream out = new DataOutputStream(s.getOutputStream());
DataInputStream in = new DataInputStream(s.getInputStream());
System.out.print("Enter first number: ");
out.writeInt(sc.nextInt());
System.out.print("Enter second number: ");
out.writeInt(sc.nextInt());
System.out.print("Enter operator (+ - * /): ");
out.writeChar(sc.next().charAt(0));
System.out.println("Answer = " + in.readInt());
s.close();
}
}

```

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```

server3.java  client3.java  ×  +
File Edit View

import java.io.*;
import java.net.*;

class server3 {
public static void main(String[] args) throws Exception {
ServerSocket ss = new ServerSocket(1234);
Socket s = ss.accept();
DataInputStream in = new DataInputStream(s.getInputStream());
DataOutputStream out = new DataOutputStream(s.getOutputStream());
int a = in.readInt(), b = in.readInt();
char op = in.readChar();
int r = (op == '+') ? a + b : (op == '-') ? a - b : (op == '*') ? a * b : a / b;
out.writeInt(r);
s.close();
ss.close();
}
}

```

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```
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C:\Admin>javac BroadcastServer.java

C:\Admin>java BroadcastServer
Sending
Sending
Sending
Sending
Sending

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C:\Admin>javac BroadcastClient.java

C:\Admin>java BroadcastClient
Message received from /192.168.50.1 Message is : This is xyz Calling.....
Message received from /192.168.50.1 Message is : This is xyz Calling.....
Message received from /192.168.50.1 Message is : This is xyz Calling.....
Message received from /192.168.50.1 Message is : This is xyz Calling.....
```

```
BroadcastClient.java X BroadcastServer.java
File Edit View

import java.net.*;
import java.io.*;

public class BroadcastClient
{
    public static final int PORT = 1234;
    public static void main(String args[]) throws Exception
    {
        MulticastSocket socket;
        DatagramPacket packet;
        InetAddress address;
        address = InetAddress.getByName("239.1.2.3");
        socket = new MulticastSocket(PORT);
        socket.joinGroup(address);
        byte[] data = new byte[100];
        packet = new DatagramPacket(data, data.length);
        for (;;)
        {
            socket.receive(packet);
            String str = new String(packet.getData());
            System.out.println("Message received from "
                + packet.getAddress()
                + " Message is : "
                + str);
        }
    }
}
```

```
BroadcastClient.java BroadcastServer.java
File Edit View

import java.net.*;
import java.io.*;
import java.util.*;

public class BroadcastServer
{
    public static final int PORT = 1234;
    public static void main(String args[]) throws Exception
    {
        MulticastSocket socket;
        DatagramPacket packet;
        InetAddress address;
        address = InetAddress.getByName("239.1.2.3");
        socket = new MulticastSocket();
        socket.joinGroup(address);
        byte[] data = null;
        for (;;)
        {
            Thread.sleep(10000);
            System.out.println("Sending ");
            String str = "This is xyz Calling.....";
            data = str.getBytes();
            packet = new DatagramPacket(
                data,
                str.length(),
                address,
                PORT
            );
            socket.send(packet);
        }
    }
}
```